

Project Management



Faculty of Applied Engineering and Urban Planning

Engineering Project Management

Defining the Project

Defining the Project

Step 1: Defining the Project Scope

Step 2: Establishing Project Priorities

Step 3: Creating the Work Breakdown Structure

Step 4: Integrating the WBS with the Organization

Step 5: Coding the WBS for the Information
System

Step 1: Defining the Project Scope

- Project Scope

- A definition of the end result or mission of the project (a product or service for the client/customer) in specific, tangible, and measurable terms.
- The scope should be developed under the direction of the project manager, customer, and other significant stakeholders.

- Purpose of the Scope Statement

- To clearly define the deliverable(s) for the end user.
- To focus the project on successful completion of its goals.
- To be used by the project owner and participants as a planning tool and for measuring project success.

Employing a Project Scope Checklist

To ensure that scope definition is complete, you may wish to use the following checklist:

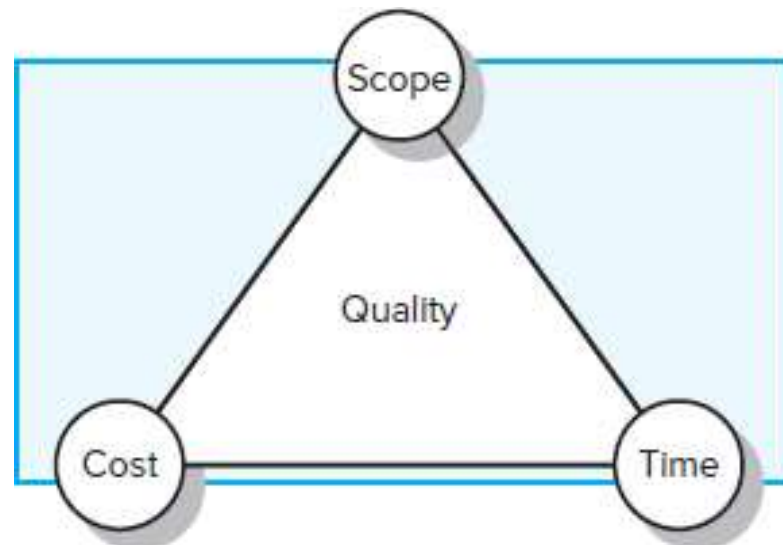
1. Project objective
2. Deliverables
3. Milestones
4. Technical requirements
5. Limits and exclusions
6. Reviews with customer



Step 2: Establishing Project Priorities

- Causes of Project Trade-offs
 - Shifts in the relative importance of criteria related to cost, time, and performance parameters
 - Budget–Cost
 - Schedule–Time
 - Performance–Scope

**Project Management
Trade-offs**



Step 2: Establishing Project Priorities

- Managing the Priorities of Project Trade-offs
 - Constrain: a parameter is a fixed requirement.
 - Enhance: optimizing a criterion over others.
 - Accept: reducing (or not meeting) a criterion requirement.

Project Priority Matrix

	Time	Performance	Cost
Constrain		●	
Enhance	●		
Accept			●

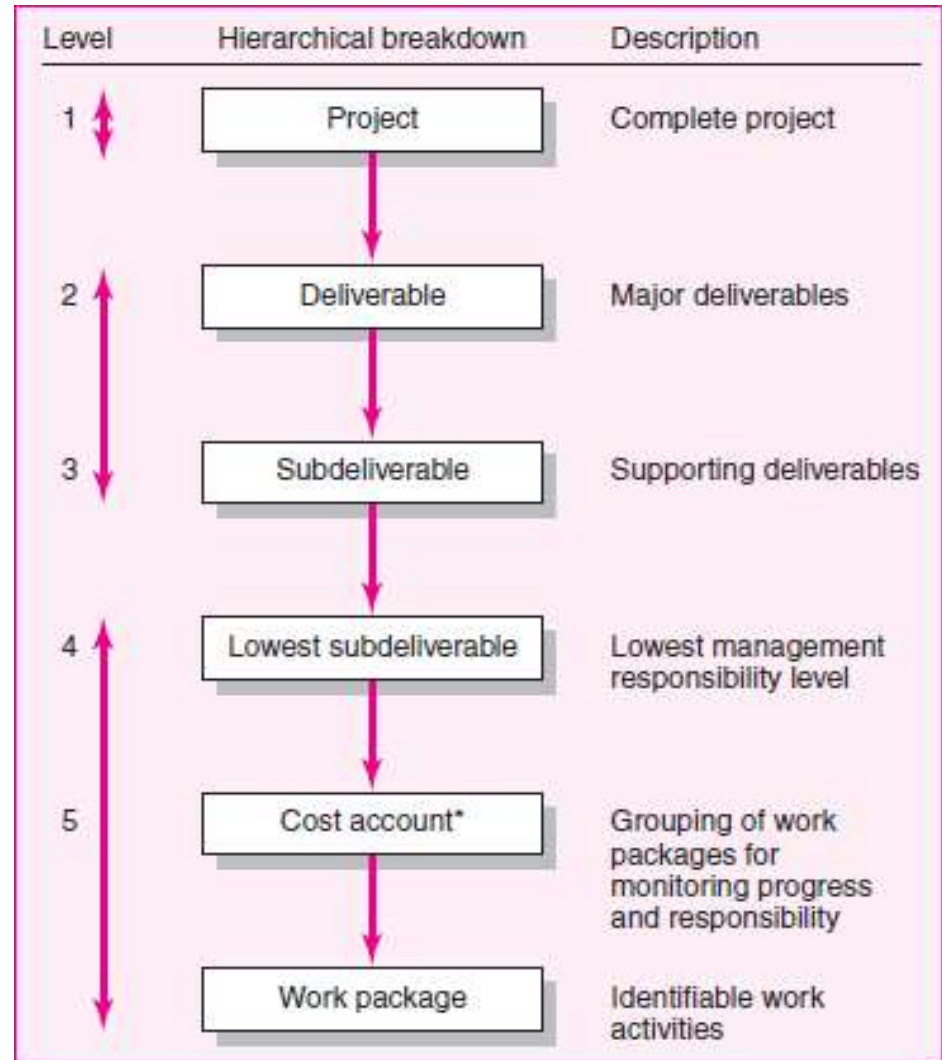
Step 3: Creating the Work Breakdown Structure (WBS)

- Work Breakdown Structure (WBS)

- ❑ WBS is described as a hierarchical structure which is designed to logically sub-divide all the work-elements of the project into a graphical presentation.
- ❑ The full scope of work for the project is placed at the top of the diagram, and then sub-divided smaller elements of work at each lower level of the breakdown.
- ❑ At the lowest level of the WBS the elements of work is called a work package.

Work Breakdown Structure (WBS)

- ❑ A list of project's activities is developed from the work packages.
- ❑ The WBS will outline the scope of the project and the responsibility for each work package.



How WBS Helps the Project Manager

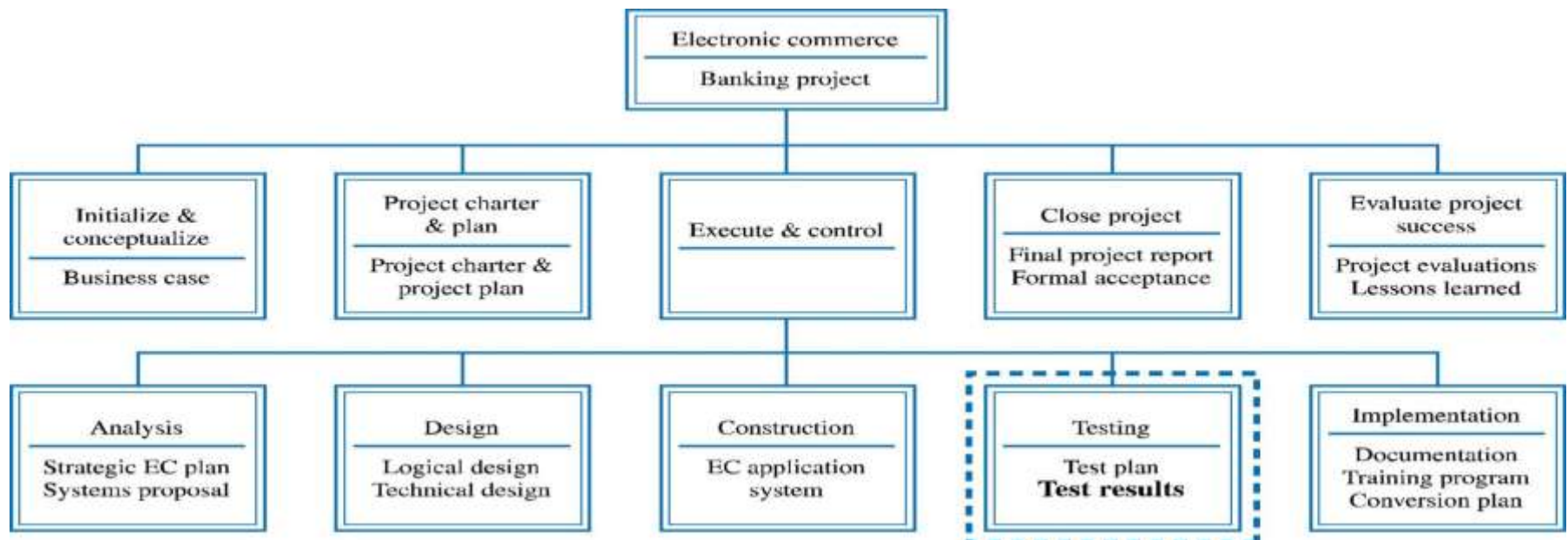
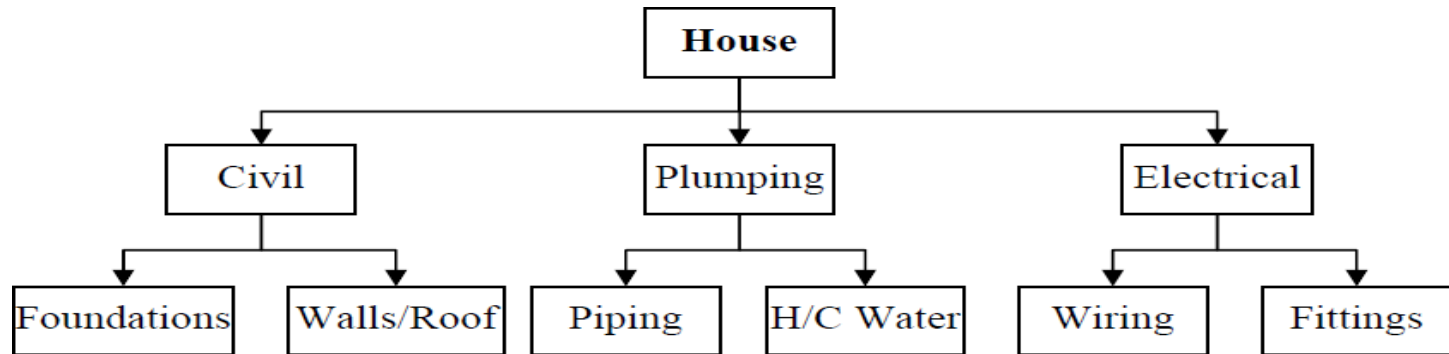
- WBS

- Facilitates evaluation of cost, time, and technical performance of the organization on a project.
- Provides management with information appropriate to each organizational level.
- Helps in the development of the organization breakdown structure (OBS). which assigns project responsibilities to organizational units and individuals
- Helps manage plan, schedule, and budget.
- Defines communication channels and assists in coordinating the various project elements.

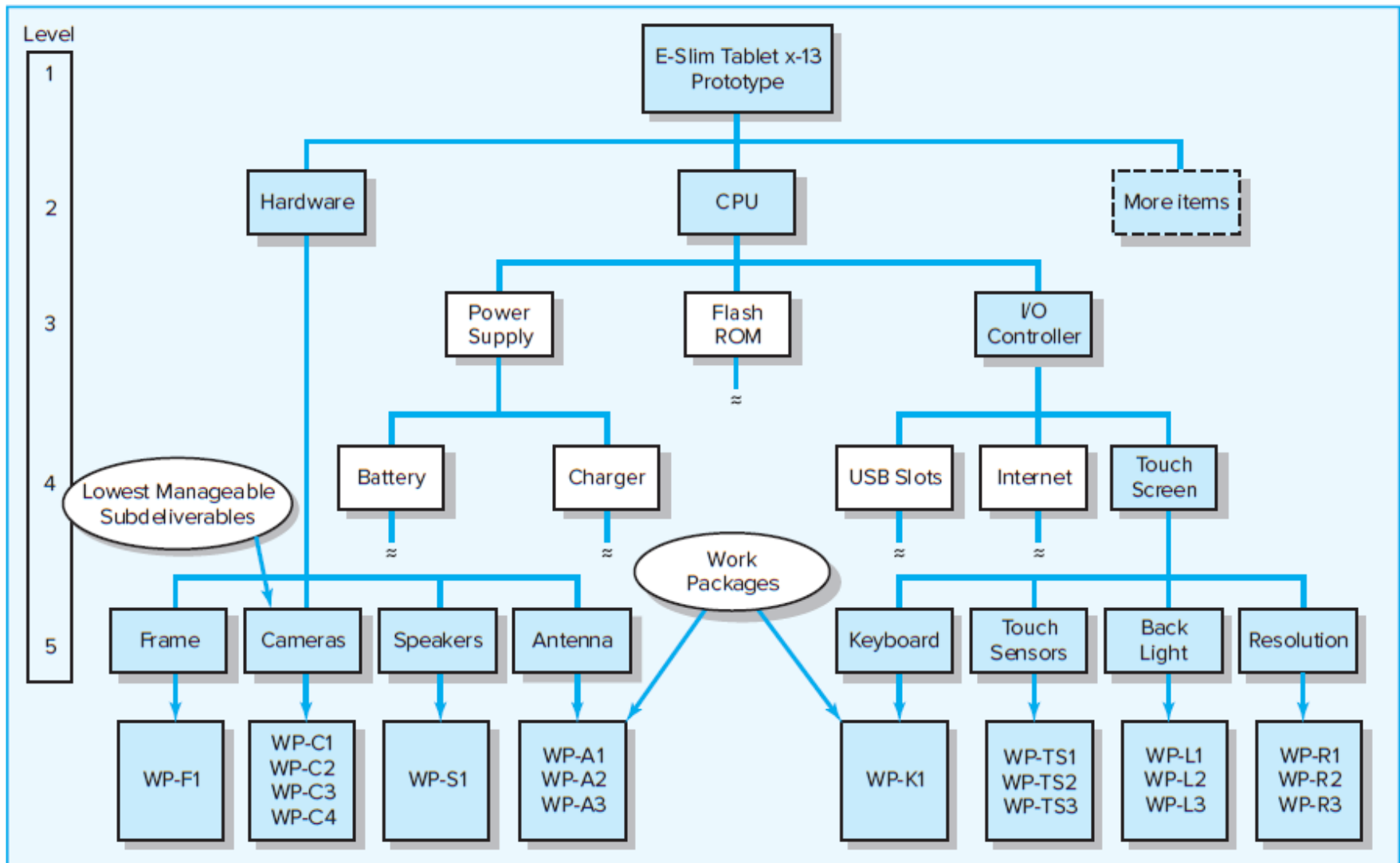
Work Packages

- A work package is the lowest level of the WBS.
 - It is output-oriented in that it:
 1. Defines work (what).
 2. Identifies time to complete a work package (how long).
 3. Identifies a time-phased budget to complete a work package (cost).
 4. Identifies resources needed to complete a work package (how much).
 5. Identifies a person responsible for units of work (who).
 6. Identifies monitoring points (milestones) for measuring success.

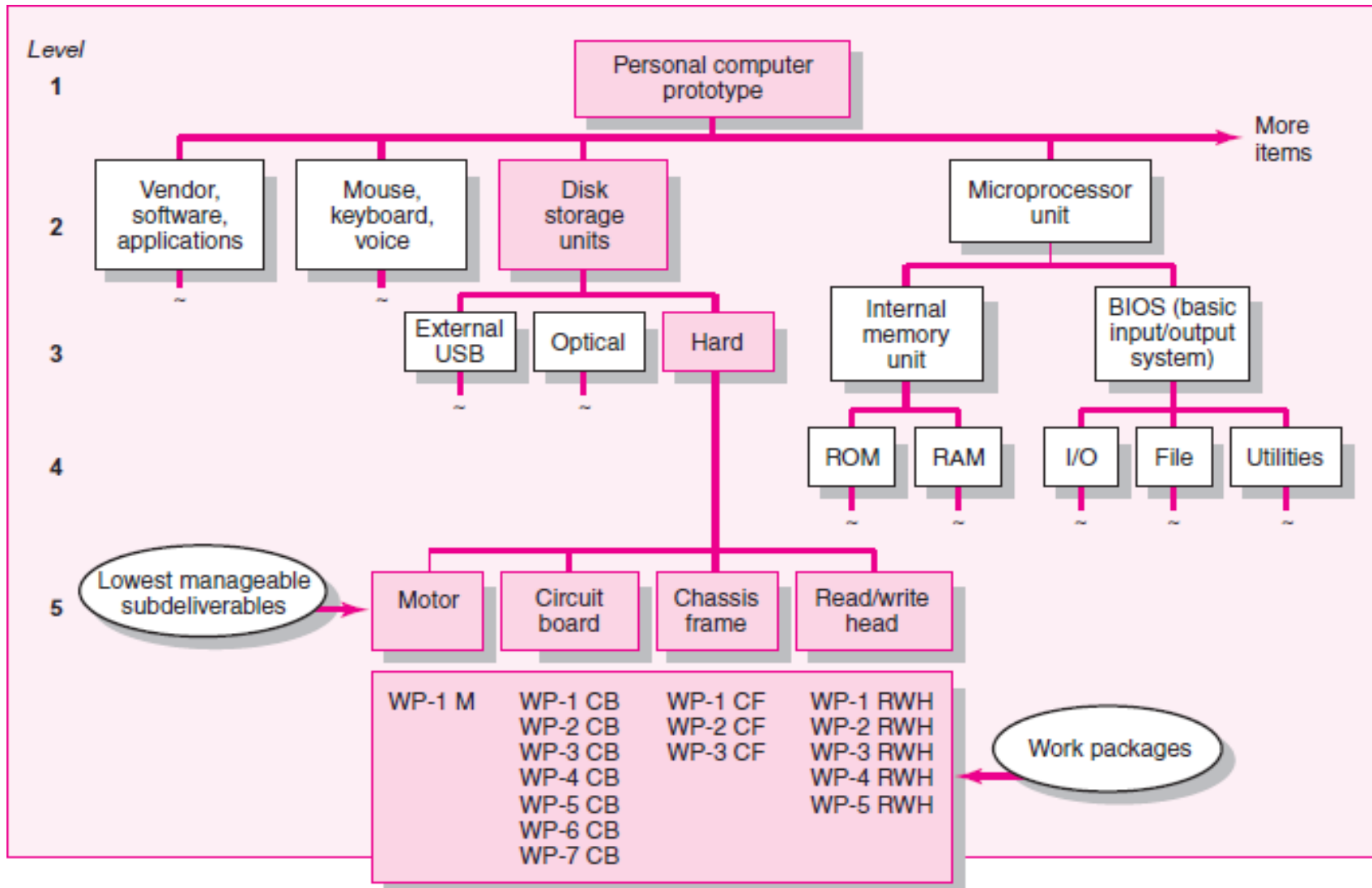
Work Breakdown Structure (WBS)



Work Breakdown Structure (WBS)



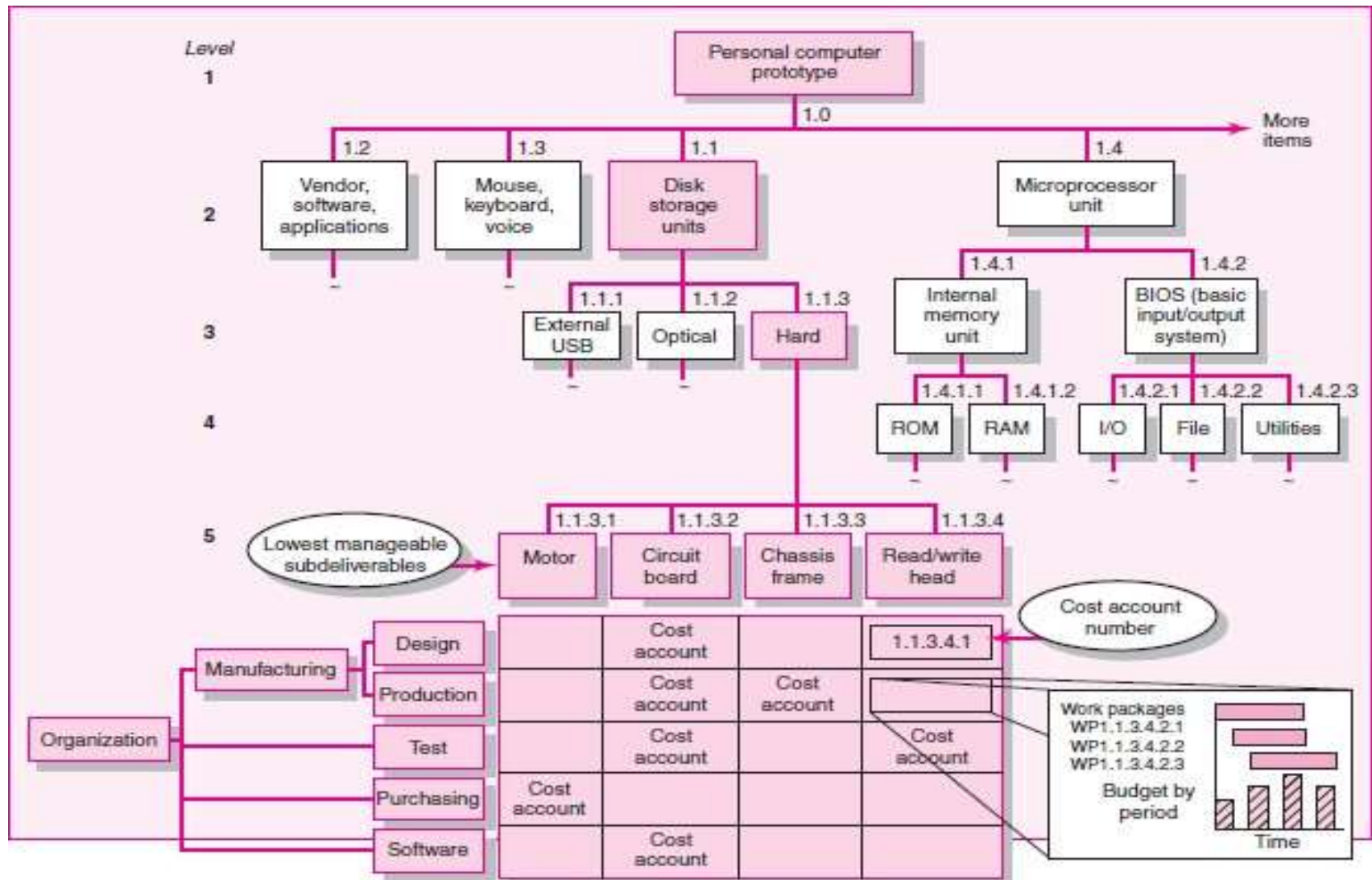
Work Breakdown Structure (WBS)



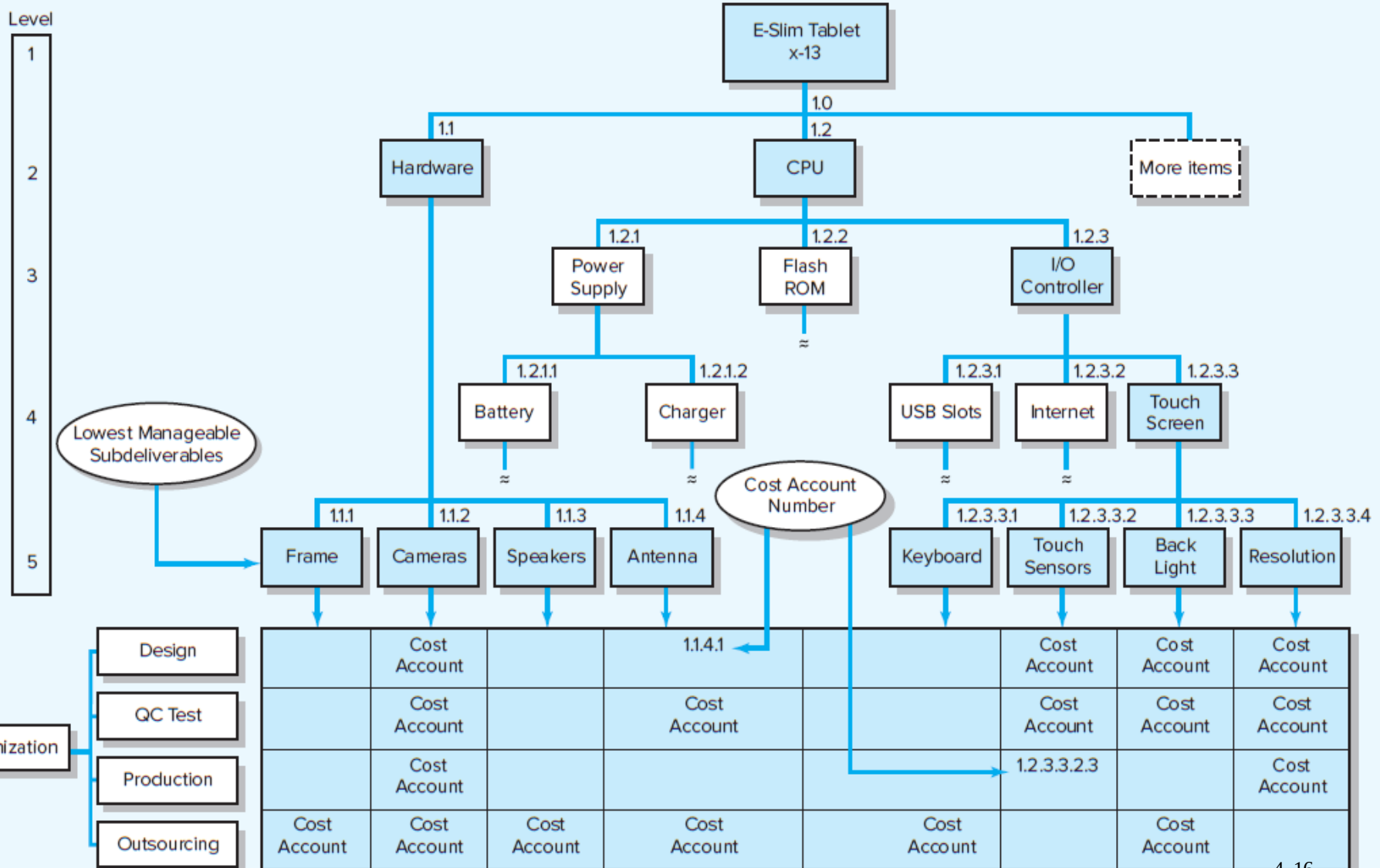
Step 4: Integrating the WBS with the Organization

- Organizational Breakdown Structure (OBS)
 - Depicts how the firm is organized to discharge its work responsibility for a project.
 - Provides a framework to summarize organization work unit performance.
 - Identifies organization units responsible for work packages.
 - Ties the organizational units to cost control accounts.

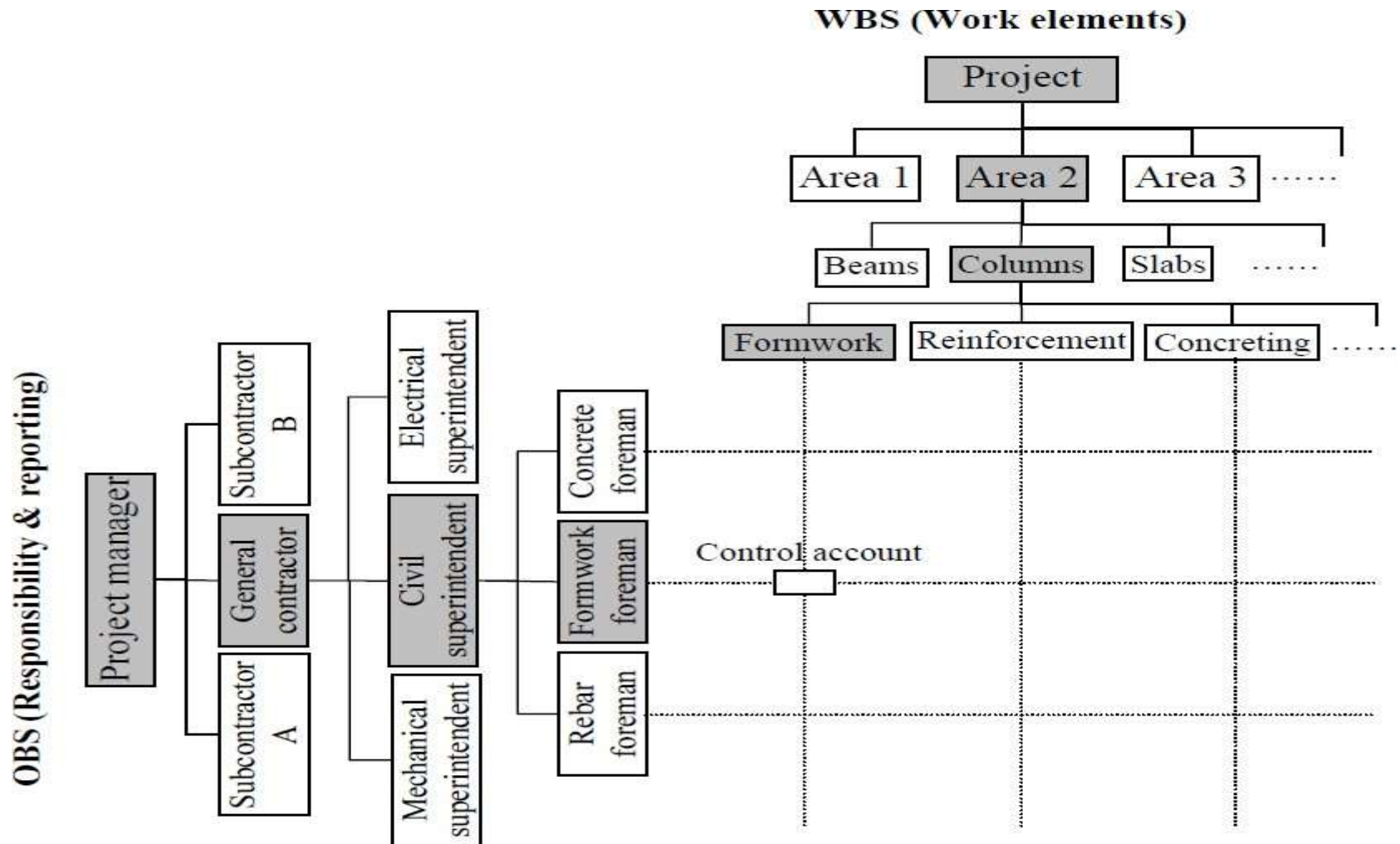
Integration of WBS and OBS



Integration of WBS and OBS



Integration of WBS and OBS



Step 5: Coding the WBS for the Information System

- WBS Coding System

- Defines:

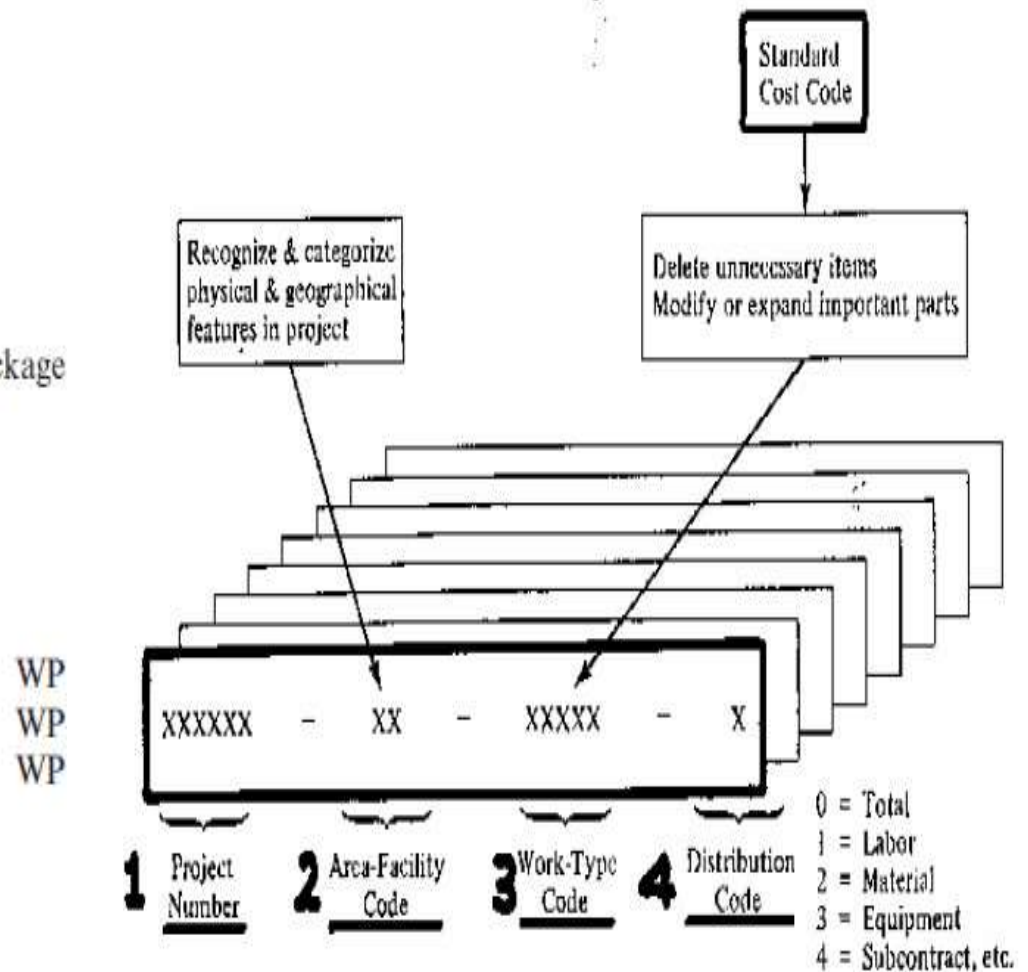
- Levels and elements of the WBS
 - Organization elements
 - Work packages
 - Budget and cost information

- Allows reports to be consolidated at any level in the organization structure



WBS Coding System

- 1.0 Computer project
 - 1.1 Disk storage units
 - 1.1.1 External USB
 - 1.1.2 Optical
 - 1.1.3 Hard
 - 1.1.3.1 Motor
 - 1.1.3.1.1 Sourcing work package
 - .
 - .
 - 1.1.3.4 Read/write head
 - 1.1.3.4.1 Cost account
 - 1.1.3.4.2 Cost account
 - 1.1.3.4.2.1 WP
 - 1.1.3.4.2.2 WP
 - 1.1.3.4.2.3 WP
 - 1.1.3.4.3 Cost account
 - .
 - .
 - .
- etc.



WBS Coding System

		Task Mode ▼	Task Name ▼
1			- 1 E-Slim Tablet x-13 Prototype
2			- 1.1 Hardware
3			1.1.1 Cameras
4			1.1.2 Speakers
5			1.1.3 Antenna
6			- 1.2 CPU
7			- 1.2.1 Power supply
8			1.2.1.1 Battery (more items)
9			1.2.1.2 Charger (more items)
10			- 1.2.2 Flash Rom (more items)
11			1.2.2.1 I/O controller
12			1.2.2.2 USB slots (more items)
13			1.2.2.3 Internet (more items)
14			- 1.2.3 Touch screen
15			- 1.2.3.1 Keyboard
16			1.2.3.1.1 Work package
17			- 1.2.3.2 Touch sensors
18			1.2.3.2.1 Work package
19			1.2.3.2.2 Work package
20			1.2.3.2.3 Work package
21			1.2.3.3 Back light (more items)
22			1.2.3.4 Resolution (more items)

Project Communication Plan

- The communication plan is usually created by the project manager and/or the project team in the early stage
- Communication is a key component in coordinating and tracking project schedules, issues, and action items.
- The plan maps out the flow of information to different stakeholders and becomes an integral part of the overall project plan.
- The purpose of a project communication plan is to express what, who, how, and when information will be transmitted to project stakeholders so schedules, issues, and action items can be tracked.

Project Communication Plan

- What information needs to be collected and when?
- Who will receive the information?
- What methods will be used to gather and store information?
- What are the limits, if any, on who has access to certain kinds of information?
- When will the information be communicated?
- How will it be communicated?

Questions?!